

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

April-May 2013

CL308 Geotechnical Engineering-II

Date: 02.05.2013, Thursday

Time: 10:00a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Give the objectives of subsurface exploration. Explain factors affecting sample disturbance. [05]
- (b) What is significant depth? [02]
- Q - 2 (a) What is meant by 'tilt' and 'shift' of a caisson? What are its causes? How can tilted caisson brought in correct position? [06]
- (b) What is caisson? With the help of neat diagram explain the classification of caisson. [04]
- (c) A soil mass is retained by a smooth backed vertical wall of 6 m height. The bulk unit weight of soil is 20kN/m^3 and $\phi=16^\circ$. The top of the soil is with the level of top of wall and is horizontal. If the soil surface carries a uniformly distributed load of 4.5kN/m^2 , determine the total active thrust on the wall and its point of application. [04]

OR

- Q - 2 (a) What is grip length of well foundation? State Lacey's formula and explain the parameters. [06]
- (b) Draw a neat sketch of well foundation and explain its various components. [04]
- (c) A vertical excavation was made in a clay deposit having unit weight of 22kN/m^3 . It caved after the digging reached 4 m depth. Assuming $\phi=0$, calculate the magnitude of cohesion of clay. [04]
- Q - 3 (a) Determine the active earth pressure on the retaining wall of 5 m high for; [05]
1. Top layer = 2.5m soil, $\phi=35^\circ$, $\gamma=17\text{kN/m}^3$
 2. Bottom layer = 2.5 m soil, $\phi=38^\circ$, $\gamma=18\text{kN/m}^3$
- Water level is at 2.5 m below ground surface. Take $\gamma_w=10\text{kN/m}^3$.
- (b) Explain with diagram vertical stress distribution on a horizontal plane due to point load acting at the ground surface. [04]
- (c) Two footings $4\text{m} \times 4\text{m}$ and $3\text{m} \times 3\text{m}$ are placed at 6m c/c apart at the same level and [05]

carry point loads 1500kN and 1250kN respectively. Compute the vertical pressure at midway between two footings at 8m depth.

OR

- Q - 3 (a) Explain Culmann's graphical method for earth pressure. [05]
 (b) Explain Newmark's influence chart. [04]
 (c) A tower of 750kN weight forms an equilateral triangle of 3m side at base and supported by footings at each corner. Calculate vertical pressure at a depth of 3m below centre of the triangle. [05]

SECTION - II

- Q - 4 (a) Explain in detail the method for locating center of critical slip circle. [05]
 (b) What is 'δ' in Swedish circle method? Write the formula for factor of safety for the same method. [02]
- Q - 5 (a) (i) Explain the types of factor of safety for the stability of slope. [03]
 (ii) Find the factor of safety of an infinite slope having angle of slope 20° . The slope is made up of cohesionless soil with value of $\phi = 35^\circ$. [02]
 (b) Explain in detail, any four factors influencing the selection of piles. [04]
 (c) A group of 9 piles of 10m long is used as the foundation of column. The piles have 25cm diameter and 90cm centre to centre distance. The sub soil consists of clay with unconfined compressive strength 170kN/m^2 . Calculate the safe load. F.O.S. = 3, adhesion factor ' α ' = 1. [05]

OR

- Q - 5 (a) Explain graphical method and rectangular plot method to calculate ΣN and ΣT . [05]
 (b) Write short note on; (i) Group action of pile, (ii) Negative skin friction. [04]
 (c) A 12m long, 300mm diameter pile is driven in a uniform deposit of sand ($\phi=40^\circ$). The water table is at a great depth and is not likely to rise. The average dry unit weight of sand is 18kN/m^3 . Using $N_q=135$ calculate the safe load capacity of the pile with F.O.S. = 2.5. Take coefficient of earth pressure ' K ' = 1.0 and value of $\delta = 30^\circ$. [05]
- Q - 6 (a) A footing 2m square is laid at a depth of 1.3m below the ground surface. Determine the net ultimate bearing capacity using IS code method. Take $\gamma=20\text{kN/m}^3$, $\phi=35^\circ$, & $C=0$. [06]

$N_c = 46.12$	$N_q = 33.30$	$N_\gamma = 48.03$
$S_c = 1.3$	$S_q = 1.2$	$S_\gamma = 0.8$

- (b) Determine net allowable load and gross allowable load for a square footing of 1.5m X 1.5m size and 1.0m depth, using Terzaghi's theory. Take $\gamma=20\text{kN/m}^3$, $\phi=25^\circ$, $C=20\text{kN/m}^2$, F.O.S. = 3.0. The values of constants are given in table below; [06]

$N_c' = 14.80$	$N_q' = 5.60$	$N_\gamma' = 3.20$
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- (c) Enlist the necessity of pile foundation.

[02]

OR

- Q - 6 (a) Using Terzaghi's theory determine the ultimate bearing capacity of a strip footing [08]

1.5m wide resting on a saturated clay ($C_u=30\text{kN/m}^2$, $\phi_u=0^\circ$ and $\gamma_{\text{sat}}=20\text{kN/m}^3$) at a depth of 2m below ground level. The water table is also at a depth of 2m from the ground level. If the water table rises by 1m, calculate percentage reduction in the ultimate bearing capacity.

$N_c = 5.7$	$N_q = 1.0$	$N_\gamma = 0$
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- (b) Define the following terms; (i) Foundation, (ii) Ultimate bearing capacity, (iii) Net safe bearing capacity, (iv) Allowable bearing pressure. [04]
- (c) Differentiate between end bearing pile and friction pile. [02]
